

Alarm Response and Escalation Procedure

VINDHYA PRECISION WORKS

STANDARD OPERATING PROCEDURE: PLANT ENGINEERING

SOP-MNT-412: Industrial Automation Alarm Response and Threshold Classification

Revision: 4.2

Effective Date: October 1, 2023

Owner: Maintenance Engineering Lead

1. PURPOSE AND SCOPE This document defines the mandatory operational thresholds, priority classifications, mandatory response timeframes, and shift-handover protocols for all machinery and facility alarms logged through the Vindhya Precision Works (VPW) SCADA and PLC monitoring architecture. Compliance is mandatory for all maintenance technicians, shift supervisors, and reliability engineers.

2. ALARM PRIORITY CLASSIFICATIONS AND RESPONSE TIMEFRAMES

All system alerts generated by the VPW instrumentation network shall be categorised into four priority tiers. Response time is defined as the elapsed duration from the timestamp of alarm annunciation in the control room to the physical arrival of a qualified technician at the asset location and the initiation of diagnostic or mitigation procedures.

* 2.1 Priority 1 (P1) — Critical Safety / Catastrophic Failure

* *Definition:* Immediate risk of personnel injury, structural asset destruction, or uncontained hazardous release.

* *Mandatory Response Time:* Under 3 minutes.

* *Immediate Action:* Automatic machine trip sequence initiates concurrently; manual verification by lead technician required before line reset.

* 2.2 Priority 2 (P2) — Major Process Deviation / Imminent Damage

* *Definition:* Operating parameters exceeding safe continuous duty thresholds; unaddressed, this will escalate to a P1 within 30 minutes.

* *Mandatory Response Time:* Under 10 minutes.

* *Immediate Action:* Controlled load reduction or manual stop initiated if parameter cannot be stabilized.

* 2.3 Priority 3 (P3) — Minor Deviation / Quality Risk

* *Definition:* Parameters drifting outside nominal operating windows; presents risk to component tolerance or product metallurgy, but no immediate asset damage.

* *Mandatory Response Time:* Within 60 minutes.

* *Immediate Action:* Secondary diagnostic check; maintenance ticket generated for next planned maintenance window.

* 2.4 Priority 4 (P4) — Informational / Maintenance Advisory

* *Definition:* Routine status changes, filter saturation warnings, or lubrication cycle prompts.

* *Mandatory Response Time:* Within 8 operating hours.

* *Immediate Action:* Logged for review during daily maintenance planning meetings.

3. SENSOR CONDITION MAPPING BY PARAMETER

Alarms shall map to priorities P1 through P4 based on real-time telemetry from installed field instrumentation according to the parameter matrices below.

3.1 Vibration Monitoring (Accelerometers / Velocity Transducers) * P1: Peak velocity exceeds 11.2 mm/s RMS (ISO 10816 Class IV rigid foundation) OR peak acceleration exceeds 4.5 g on high-speed spindle bearings. * P2: Velocity between 7.1 mm/s RMS and 11.1 mm/s RMS for a duration exceeding 15 continuous seconds. * P3: Velocity between 4.5 mm/s RMS and 7.0 mm/s RMS . * P4: Velocity between 2.8 mm/s RMS and 4.4 mm/s RMS (trend tracking advisory).

3.2 Temperature Monitoring (RTDs / Thermocouples) * P1: Winding or bearing temperature exceeds $+95^{\circ}\text{C}$ on Class F insulation motors, OR process media temperature exceeds design limit by $+15^{\circ}\text{C}$. * P2: Temperature between $+85^{\circ}\text{C}$ and $+94.9^{\circ}\text{C}$ on bearings, or rate of rise exceeds $+3.0^{\circ}\text{C}/\text{minute}$. * P3: Temperature between $+75^{\circ}\text{C}$ and $+84.9^{\circ}\text{C}$. * P4: Low-temperature warning (fluid systems below $+15^{\circ}\text{C}$ operating threshold).

3.3 Pressure Monitoring (Strain-Gauge Transducers) * P1: Hydraulic system pressure exceeds 240 bar (relief valve backup failure state) OR drops below 12 bar on critical lubrication headers. * P2: System pressure deviates by $\pm 15\%$ from nominal setpoint for more than 45 seconds. * P3: System pressure deviates by $\pm 10\%$ from nominal setpoint. * P4: Filter differential pressure drop reaches $\geq 1.2 \text{ bar}$ (maintenance cleaning advisory).

3.4 Electrical Current Monitoring (Hall Effect / CT Sensors) * P1: Motor current draw exceeds 130% of Full Load Amps (FLA) for $>3.0 \text{ seconds}$, or phase imbalance exceeds 15% . * P2: Current draw between 115% and 129.9% FLA, or phase imbalance between 10% and 14.9% . * P3: Current draw between 105% and 114.9% FLA. * P4: Idle current drop below no-load baseline (indicates mechanical decoupling or belt slip).

4. SUSPECTED FAULTY SENSOR PROTOCOL

When sensor output displays erratic fluctuations, flat-lining, out-of-range negative values, or disagrees entirely with mechanical secondary gauges, the sensor shall be classified as suspected faulty.

* 4.1 Mandatory Restriction: A suspected-faulty sensor may never be suppressed, bypassed, or masked in the SCADA/PLC logic without an active, approved work order generated in the VPW Computerized Maintenance Management System (CMMS).

* 4.2 Verification Duty: The responding technician must perform a physical validation check using calibrated handheld diagnostic meters (e.g., Fluke multimeter, digital pressure gauge, or optical tachometer) within the time frame specified for a P3 alarm.

* 4.3 Temporary Action: If sensor failure is verified, the technician shall log the CMMS work order, switch the affected control loop to manual operation under direct supervision of the Shift Production Supervisor, and affix a physical "Out of Calibration / Faulty" tag to the local junction box within 2 hours of discovery.

5. SHIFT-HANDOVER REQUIREMENT FOR OPEN ALARMS

Continuity of plant safety and asset protection requires strict tracking of unacknowledged or active alarms across shift changes.

* 5.1 Handover Log: Thirty (30) minutes prior to the end of each shift, the outgoing Maintenance Lead and incoming Maintenance Lead shall jointly review the Master Alarm Summary Screen in the central control room.

* 5.2 Open Alarm Accounting: Every P1, P2, and P3 alarm remaining in an active or acknowledged-unresolved state must be formally transcribed into the Shift Handover Ledger, detailing:

- * Alarm ID and Asset Tag Number

- * Timestamp of initial occurrence

- * Current diagnostic status and assigned technician

- * CMMS work order reference number

* 5.3 Sign-Off: Both outgoing and incoming Maintenance Leads must digitally sign the Shift Handover Ledger. Unsigned open alarms shall constitute a breach of shift safety protocol, reportable to the Plant Operations Manager.